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Geog 450

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Visualization of Departure Flights from SeaTac Airport in 2024

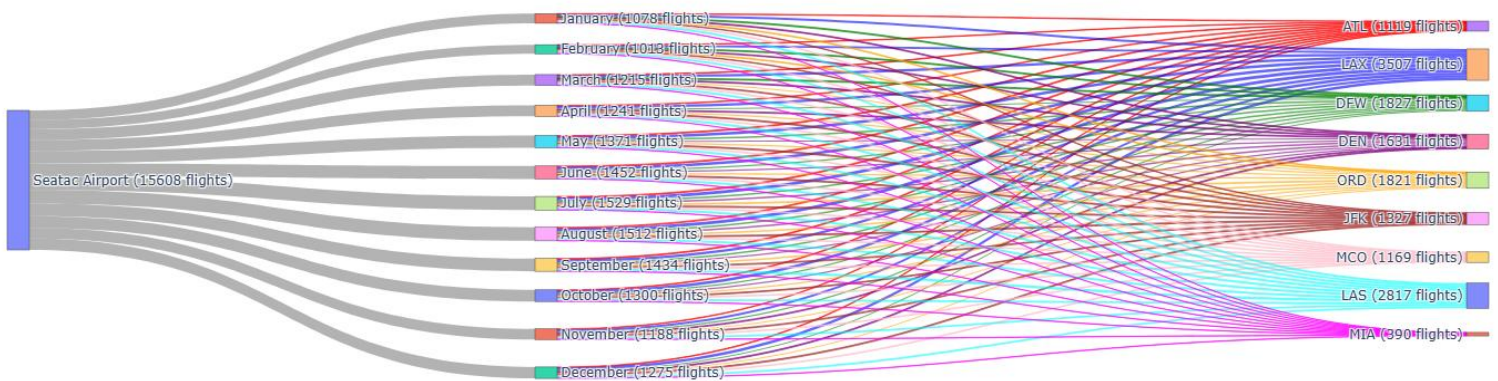
In this project we were tasked with displaying a movement-based topic in Seattle through a Sankey Diagram. We landed on airline flights from Seattle - Tacoma Airport, also referred to as SEA or SeaTac. The dominant airport of the US Pacific Northwest, the airport contributes a year-round circulation of goods and people through Seattle. We were eager to see what one year of flight data through SeaTac looks like.

We found our source in the Bureau of Transportation Statistics' data portal (link to portal: [Bureau of Transportation Statistics - Departures](#)). We decided to plot Departure flights flown by Alaska Airlines from SeaTac to other domestic airports in 2024. Alaska Airlines is the most popular airline at SeaTac, comprising over 50% of total flights through SeaTac. They get dedicated Arrival and Departure loading bays for pick up and drop off car traffic. Our key metrics were the monthly outbound flights from SeaTac, the common destination airports, and the number of flights to each destination. We used the most recent complete year for the dataset, which was 2024.

Alongside our dataset we read two articles to influence our diagram design. The first comes from the Port of Seattle, the organization who runs SeaTac Airport. [The article](#) was written in August of last year and discusses the record number of travelers through SeaTac in 2025. We learned that June, July, and August should have the highest numbers of flights. We used the second article, written by [Travel Pander](#) in July 2025, in combination with our dataset to select the most frequented domestic airports for our diagram. We chose the following: Atlanta ATL; Denver DEN; Dallas DFW; New York City JFK; Las Vegas LAS; Los Angeles LAX; Orlando MCO; Miami MIA; Chicago ORD.

Fig 1: Sankey Diagram Alaska Airlines Flights from SeaTac (plotted 2026)

Flow of Alaska Airlines flights each month from Seatac airport to destination airport in 2024



This is our Sankey diagram, a left to right flowchart that shows the movements and changes of subjects passing through different nodes. The lines represent the subjects, diverging and converging as appropriate. The nodes themselves scale in size based on the associated value. Larger nodes represent high numbers of flights and vice versa. The nodes for each month are chronologically ordered, but the nodes for each are not ordered.

Alaska Airlines flew between 1013 flights in February to 1529 flights in July. The values for destination flights are spread more greatly, ranging from 390 flights for Miami MIA to 3507 flights for Los Angeles LAX. The article from Travel Pander details more about the travelers at each airport. Airports like LAX, DEN, and LAS host large numbers of tourists seeking winter recreation, warm landscapes and vibrant culture. Airports like JFK, ORD, and ATL handle more business and corporate traffic. This isn't a binary distinction, each airport will admit a variety of travelers with different goals.

The relative consistency of flights per month shows that Alaska Airlines maintains a strong connection between Seattle and the rest of the country. These nine airports span the width of the US. There also seems to be a preference for warmer climate. Six of the airports are in the southern US, the other three located central and in the north.

Our Sankey diagram gives a first impression of travel patterns out of Seattle. I expected to see more winter travel considering the December holidays. But the diagram suggests summer remains more practical for most flyers. I also underestimated the wide range of destinations commonly serviced by Alaska Airlines.

Our diagram was compromised by time and resource constraints. We restricted our scope to Departure flights and basic travel information, and I feel the diagram lacks the full

potential of a Sankey diagram. The diagram makes the data more interesting but reveals fewer patterns than I'd hoped. The basic conclusions are that Alaska regularly operates from SeaTac each month, and they service these nine airports every month.

We omitted Arrivals information for resource and time reasons, but in the future I'd want to incorporate that information. It would also be nice to show information on cargo versus passenger traffic. These details would add color and perspective to the flow of goods and people through Seattle.